Allison Brown

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EDUCATION

Florida Gulf Coast University | Fort Myers, FL — Expected May 2027

B.S. Software Engineering | GPA: 3.76 | Dean's List 4× | President's List 2× | First-Generation Student

Relevant Coursework: Data Structures & Algorithms, Computer Security, Systems Administration & Programming, Operating Systems, Cloud Computing, Requirements Engineering, Data Engineering

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, HTML/CSS, Bash

Frameworks & Tools: React Native, FastAPI, Playwright, Selenium, TestNG, Azure Functions, Google Cloud, Red Hat Linux, Git

Concepts: REST API Design, Event-Driven Architecture, Machine Learning, Computer Vision, Automated Testing, Agile/Scrum

EXPERIENCE

Student Intern - *Homeland Security Investigations (HSI)* | Fort Myers, FL — January – August 2026

- Conducted digital forensic analysis on mobile device evidence across 10+ federal cases under strict chain-of-custody protocols.
- Collaborated with federal investigators to scope and deliver technical workflows in a high-compliance, multi-agency environment.
- Built a Python/FastAPI forensic media hashing prototype projected to reduce evidence categorization time by ~20% if deployed.

Software Developer - *BoomBox Inc.* | Fort Myers, FL — September 2025 – July 2026

- Developed React Native components for a cross-platform music discovery app, integrating 8+ REST APIs to enable data-driven features.
- Built and redesigned the onboarding flow and settings page, shipping light/dark mode and Apple Music integration within sprint deadlines.
- Collaborated with 12+ developers in a shared GitHub Codespace, scoping and delivering features on time across structured agile sprints.

Outreach Assistant - *FGCU U.A. Whitaker College of Engineering* | Fort Myers, FL — September 2026 – Present

- Lead planning for the h{AI}ckathon at the college's AI Day (November 2026), delegating and managing work across a 2-person team.

PROJECTS

Traffic & Pavement Intelligence Backend - *Python, Azure Functions, ML, REST APIs, Digital Twin* — August – December 2025

- Built an event-driven Azure backend ingesting live FDOT sensor data into Digital Twin road-segment models using 7 Azure Functions.
- Trained ML models to forecast pavement conditions and traffic trends, exposed via REST APIs for downstream reporting systems.

Drone Security Simulations - *Python, ArduPilot SITL, MAVLink, Gazebo, MAVProxy* — February – April 2025

- Simulated RF jamming attacks on delivery UAVs across 3 adversarial scenarios in ArduPilot SITL + Gazebo to find comms vulnerabilities.
- Developed a Python frequency-hopping defense that dynamically rotates channels to evade active jamming, validated in simulation.
- Programmed a MAVLink heartbeat monitor with a 5-second failsafe that autonomously issues LAND commands on signal loss.

iMPOSTURE: AI-Powered Posture Coach - *Python, MediaPipe, YOLOv11, Google ADK, Cloud Logging* — September 2025

- Built a real-time computer vision pipeline with MediaPipe and YOLOv11 for posture and phone detection in a 4-person team in 24 hours.
- Deployed 3 Google ADK agents for posture, distraction, and noise monitoring, streaming 38K+ rows to Google Cloud Logging.
- Placed Top 11 overall and 2nd in the Google Cloud ADK Challenge among 1,400+ hackers at ShellHacks 2025 (FIU).

DisasterAssistance.gov: Automated Test Suite - *Java, Selenium WebDriver, TestNG, Maven* — March 2026

- Built a Selenium/TestNG suite with 8 test classes and 40 tests validating key user flows on a live US government emergency-resource site.
- Designed an OOP test architecture (BaseTest inheritance, testng.xml sequencing), eliminating boilerplate across all 40 tests.

LEADERSHIP & ACTIVITIES

Founding President - *INIT FGCU Chapter* — May 2026 – Present

- Founded the FGCU chapter of a national tech organization, expanding access to national industry partnerships and career opportunities.
- Lead the executive board, chapter strategy, and technical programming to drive organizational growth and engagement.

Founder & Lead Organizer - *FGCU EagleHacks* — April 2024 – Present

- Founded and led 3 annual iterations of FGCU's hackathon, growing attendance to 100+ participants with $23K+ from 7 sponsors.
- Coordinated with university administration and campus police to secure overnight access and open the event to outside participants.

President - *FGCU Computer Science & Software Engineering Club (CSSEC)* — January 2024 – Present

- Promoted from Marketing Coordinator to President, leading a 100+ member organization with 12–14 workshops and panels per semester.
- Organized career events featuring Microsoft, Hertz, Arthrex, and more, consistently engaging 50–60 students per session.